COMP 4911
Developing Enterprise Service

School	School of Computing and Academic Studies
Program	Computer Systems Technology (CST) Diploma
Course Credits	5
Minimum Passing Grade	50%
Start Date	January 05, 2026
End Date	April 24, 2026
Total Hours*	75
Total Weeks	15
Hours/Weeks	5
Delivery Type	Lecture/Lab
Prerequisite(s)	• 50% in COMP 3910
CRN	19461

Acknowledgement of Territories

The British Columbia Institute of Technology acknowledges that our campuses are located on the unceded traditional territories of the Coast Salish Nations of Sḵwx̱wú7mesh [1] (Squamish), səl̓ilwətaʔɬ [2] (Tsleil-Waututh), and xwməθkwəy̓əm [3] (Musqueam).

Instructor Details

Name	Bruce Link
E-mail	Bruce_Link@bcit.ca
Location	SW2-365
Office Hours	See Learning Hub welcome message, by appointment (email instructor).

Course Description

This course completes the overall Information Technology (IT) framework introduced in Comp 3910, and offers the student the opportunity to apply IT skills in a realistic work environment. In particular, this course will focus on: Information systems planning and management; Project management for medium and large teams; Software Development Lifecycle (SDLC) issues; Component technology using Enterprise Java Beans (EJBs); Advanced Java, Web component, Java EE, .NET, Visual Studio, Microsoft BackOffice and/or Microsoft Office Technologies, as appropriate.

Course Learning Outcomes/Competencies

Upon successful completion of this course, the student will be able to:

- Work in a structured, team environment.
    - Use the Unified Process in a team environment.
    - Produce the artifacts required by the Unified Process.
    - Experience the principle roles defined by the Unified Process in a team environment.
    - Make and keep commitments.
    - Plan and track software development activities.
    - Organize and operate day to day IS operations.
    - Participate in peer assessments and structured performance reviews.
- Program effectively with advanced techniques such as:
    - JSF, Facelets, JSP
    - Java EE EJBs
    - CDI
    - Microsoft's .NET
    - MySQL or SQL Server databases
- Incorporate multi-tier development strategies in transaction processing systems, such as:
    - 2-tier, 3-tier, multi-tier
    - Data access methodologies, Object to Relational Mapping
    - Distributed Objects (Java EE)

Learning Resources

Required:

- Enterprise JavaBeans 3.1, Rubinger, Andrew Lee and Bill Burke, Sixth Edition, O'Reilly ISBN 978-0-596-15802-6, September 2010.

Recommended:

- The Unified Software Development Process, Ivar Jacobson, Grady Booch, James Rumbaugh, Addison-Wesley, ISBN 0-201-57169-2, 1998
- The Rational Unified Process - An Introduction, by Philippe Kruchten, Addison-Wesley, 1998

Evaluation Criteria

Criteria	%	Comments
Project artifacts	35	
Peer assessment	10	
Quizzes	10	
Midterm	20	
Final	25	comprehensive

Attendance Requirements

Regular attendance in lectures, seminars, and labs is seen as critical to student success, and may be monitored by faculty.

Unapproved absense of 2 or more classes may result in failure or forced withdrawal from the course or program.

Please see Policy 5101 — Student Regulations: https://www.bcit.ca/files/pdf/policies/5101.pdf [4]

Other information

No student recording (except accommodations)

Recording in this course: Recording is not permitted by students unless required under Policy 4501 (Accommodation for Students with Disabilities). Where an approved accommodation includes recording, Accessibility Services will notify the instructor via the IAP. Any other use or sharing of recordings is not allowed beyond the original instructional purpose.

Course Schedule and Assignments

Week:	Date	Topic	Readings	Lab
1	Jan. 5	The Comp 4911 project SW best practices Intro to Earmed Value	EJB Ch.1, 2 Job Applications by the end of the week	

2	Jan. 12	Introduction to RUP, Inception Use Case Driven Architecture Centric Java EE resources First EJB	EJB Ch 3, 4	Form teams and plan project.
3	Jan. 19	UP Iterations Stateless Session Beans Stateful Session Beans	EJB Ch 5,6 https://docs.jboss.org/weld/reference/latest/en-US/html/index.html	**Milestone: Inception**
4	Jan. 26	Requirements Capture Singleton Session Bean Entity Manager	EJB Ch 7,9	
5	Feb 2	Requirements as Use cases Mapping Persistent Objects Entity Inheritance	EJB Ch 10,12	**Milestone: Elaboration Iteration 1**
6	Feb. 9	Analysis Entity Relationships	EJB Ch 11	Peer Performance Appraisal One
7	Feb 16	Persistence Queries, and Query Language	EJB Ch 13	**Milestone: Elaboration Iteration 2**
8	Feb. 23	**Midterm Exam Week** – Non-Instructional Week		
9	Mar 2	Design Entity Callbacks and Listeners EJB Security	EJB Ch 14, 15	
10	Mar. 9	Implementation and Integration Message-driven Beans **In-class Instructional Assessments Online**	EJB Ch 8	**Milestone: Construction Iteration 1** Peer Performance Appraisal Two
11	Mar. 16	**Computing Reading Week** – Non-Instructional Week		
12	Mar. 23	Testing JNDI, ENC, Injection	EJB Ch 16	

13	Mar. 30	EJB Transactions	EJB Ch 17	
14	Apr. 6	UP General workflow **or** Java EE design	EJB 18, 19 or notes	**Milestone: Construction iteration 2** Peer Performance Appraisal Three
15	Apr. 13	**Study/Makeup/Review/Presentation Week**		
16	Apr. 20	**Final Exam Week** – Non-Instructional Week		

Course topics

BCIT Policy

The following statements are in accordance with the BCIT Policies 5101, 4501, 5103, 5104, and 7507, and their accompanying procedures. To review these policies and procedures please click on the links below.

Attendance
In case of illness or other unavoidable cause of absence, the student must communicate as soon as possible with their instructor or Program/Department Head, stating the reason for the absence. When absences result in missed safety requirements, exams, or other deadlines, an instructor or Program/Department Head may request appropriate supporting documentation, including a medical note. For all other absences, the student is responsible to ensure they seek out missed information, preferably from a classmate or recording (when available). Please see BCIT Policy 5101 - Student Regulations [5] , and accompanying procedures, for more information.

Attempts
As stated in BCIT Policy 5103 - Student Evaluation [6] , students must successfully complete a course within a maximum of three (3) attempts at the course. Students with two attempts in a single course will be allowed to repeat the course only upon permission from the Associate Dean. Students who have not successfully completed a course within three attempts will not be eligible to graduate from their respective program. For those courses or programs that have Education Council approval, the number of attempts as stated in the evaluation section of the course outline shall apply.

Academic Integrity
It is the responsibility of all students to be familiar with the Student Code of Academic Integrity. Violations of the Code, including plagiarism, cheating, misrepresentation, and academic advantage, are prohibited and will be handled in accordance with BCIT Policy 5104 – Student Code of Academic Integrity [7] , and accompanying procedures.

Accommodation
Any student who may require accommodation from BCIT because of a physical or mental disability should refer to BCIT's Policy 4501 - Accommodation for Students with Disabilities [8] , and contact BCIT's Accessibility Services (SW1 2360, 604-451-6963) at the earliest possible time. Requests for accommodation must be made to Accessibility Services, and should not be made to a course instructor or Program area.

Any student who needs special assistance in the event of a medical emergency or building evacuation (either because of a disability or for any other reason) should promptly inform their course instructor(s) and Accessibility Services of their personal circumstances.

Human Rights, Harassment and Discrimination
The BCIT community is made up of individuals from every ability, background, experience and identity, each contributing uniquely to the richness and diversity of the BCIT community as a whole. In recognition of this, and the intrinsic value of our diversity, BCIT seeks to foster a climate of collaboration, understanding and mutual respect between all members of the community and ensure an inclusive accessible working and learning environment where everyone can succeed. Respect, Diversity, and Inclusion [9] is a supportive resource for both students and employees of BCIT, to foster a respectful learning and working environment. Any student who feels that they are experiencing discrimination or harassment (personal or human rights-related) can confidentially access this resource for advice and support. Please see BCIT Policy 7507 – Harassment and Discrimination [10] and accompanying procedure.

Students should make themselves aware of additional Education, Administration, Safety and other BCIT policies listed at https://www.bcit.ca/about/administration/policies.shtml [11]

Guidelines for School of Computing and Academic Studies

No school specific policies, please refer to main BCIT Policy.

Approved

I verify that the content of this course outline is current.
Bruce Link, Faculty
December 30, 2025

I verify that this course outline has been reviewed.
Chi En Huang, Program Head
January 02, 2026

I verify that this course outline has been reviewed and complies with BCIT policy.
Donna Turner, Associate Dean
January 05, 2026

Note: Students will be given reasonable notice if changes are required to the content of this course outline.

*Course hours and credits are calculated per Policy 5012 [12] and the associated procedure [13] .

Total hours – Example of 3 credit lecture/lab course:

- ***Full-time course****: 45 hours of scheduled learning*
- ***Flexible Learning course****: 36 hours of scheduled learning plus 9 hours of independent (non-scheduled, non-instructional) learning*

List of links found on this page

This list includes all links found on this page for your reference.

[1] https://www.squamish.net/
[2] https://twnation.ca/
[3] https://www.musqueam.bc.ca/
[4] https://www.bcit.ca/files/pdf/policies/5101.pdf
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